

Assignment 3 — AI Pipeline

Student: Sayyam Shahid

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Internship: AI Automation Institute

University: FUUAST, Islamabad

AI Pipeline Explanation

The purpose of this project is to develop a **Multi-Input AI Analysis Pipeline** that can automatically process different types of information and generate useful AI-based results. The problem solved by this pipeline is that users normally have to manually open different files, read their contents, analyze the information, and prepare reports. This process can take significant time, especially when many files are involved. The developed pipeline combines these activities into one automated workflow.

The pipeline accepts multiple types of input, including **Text files, CSV files, PDF files, Images, JSON files, Markdown files, and direct User Questions**. Python first detects the input type by checking the file extension. Different processing functions are then used according to the type of input. Text and Markdown files are read directly. CSV files are read using Python's CSV module and converted into structured JSON text. JSON files are loaded and formatted. PDF documents are processed using the **PyPDF2** library to extract their text. Images are passed directly to the AI model for visual analysis.

After the input has been processed, Python prepares a structured prompt for the AI model. The prompt asks the AI to provide an **Executive Summary, Main Information, Important Points, Key Findings, Data Analysis where applicable, Questions and Answers, and Conclusion**.

The AI model used in this project is the **Gemini API**, accessed through Google's Python Gen AI SDK. The program creates a Gemini client using an API key stored in the .env file and sends the prepared information to the model. For image inputs, the image is sent to Gemini as a multimodal input.

The final output is generated as an **AI analysis report**. For file inputs, the program saves the result in both **TXT and JSON formats**. For user questions, it saves the question together with the AI answer in a separate text file.

During development, some limitations can occur. The Gemini API requires a valid API key, model availability can change, PDF text extraction may not work perfectly for scanned documents, and AI results depend on the quality of the provided input. The program also includes a retry mechanism for temporary Gemini API failures.

Pipeline Diagram

